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TEC–402

B. TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING)

(FOURTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

ANALOG CIRCUITS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the working of differential amplifier. Derive expression of I_{CQ} and V_{CEQ} for the DC analysis of differential amplifier. (CO1)

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OR

(b) Explain the internal block diagram of IC 741. Explain current mirror circuit. (CO1)

2. (a) Derive the expressions for differential mode gain and common mode gain for differential amplifier. (CO1)

OR

(b) A differential amplifier has inputs $V_1 = 7$ mV and $V_2 = 9$ mV. It has a differential mode gain of 80 dB and CMRR of 90 dB. Find the output voltage. (CO1)

3. (a) Explain the working of log and anti-log amplifier with the help of a neat diagram. (CO2)

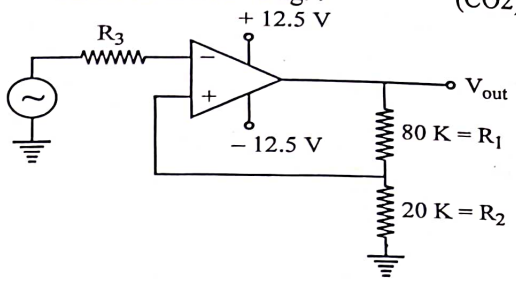
OR

(b) Design integrator, differentiator and subtractor using operational amplifier. (CO2)

4. (a) Derive expression for first order high pass filter. Draw and explain the frequency response curve of band pass and band reject types of filters. (CO2)

OR

(b) Determine the threshold voltages and the hysteresis voltage of Schmidt trigger circuit as shown in fig. : (CO2)



5. (a) Draw different configurations of differential amplifier. Explain the circuit of level shifter. (CO1, CO2)

OR

(b) Explain the working of square and triangular wave generator with the help of a diagram. (CO1, CO2)